

# Deterministic Optimization

Linear Optimization Modeling

Decision-Making under Uncertainty

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A Concrete Example

# Modeling using Linear Programs

## Learning Objectives for this Lesson

- Introduce scenario tree  
representation of uncertainty
- Solve a concrete example of two-  
stage stochastic program
- Analyze the results

# Stochastic Inventory Control

- Decision variables:
  - **Here-and-Now decision:** production quantity  $q$
  - **Wait-and-See decision:** Sell quantity  $y$ , Discount (savage) quantity  $v$
- Objective: minimize production cost and expected future cost
- Stochastic program:

$$\begin{aligned} & \min_x cx + E_d[Q(x, d)] \\ & \text{s.t. } 0 \leq x \leq \bar{x} \end{aligned}$$

} **First-Stage** Problem with  $q$  as first-stage decision

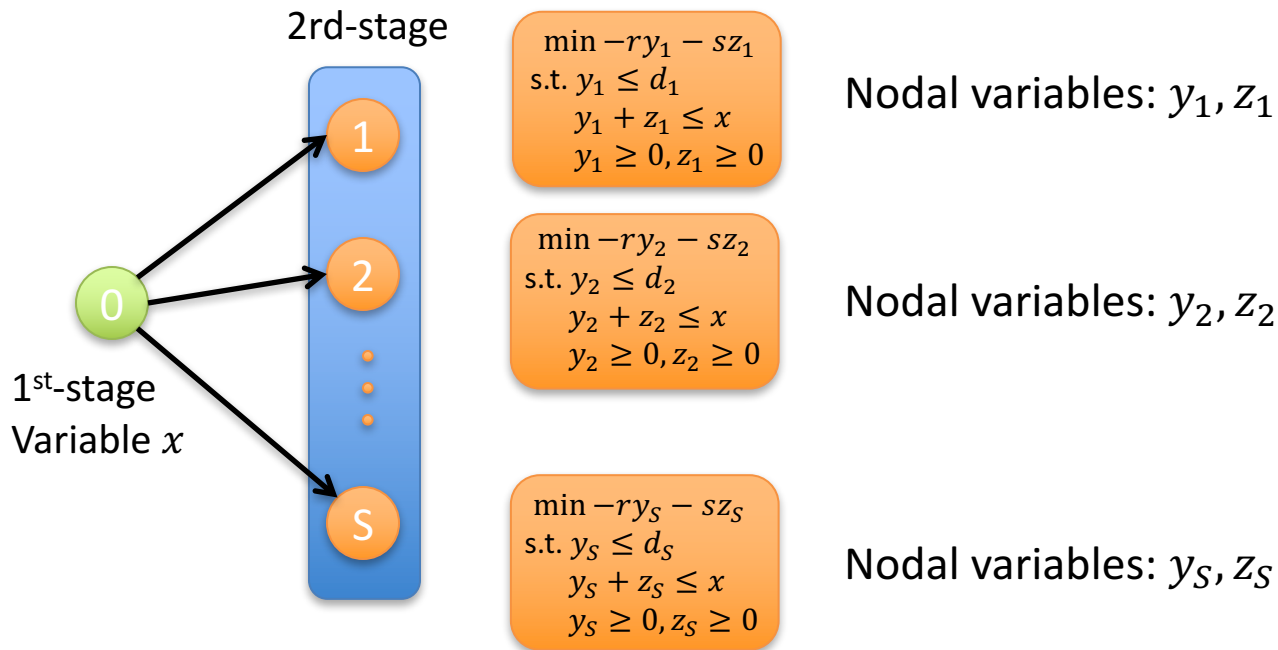
$$\begin{aligned} & Q(x, d) = \min_{y, z} -r \cdot y - s \cdot z \\ & \text{s.t. } y \leq d \\ & \quad y + z \leq x \\ & \quad y \geq 0, z \geq 0 \end{aligned}$$

} **Second-Stage** Problem with second-stage decision  $y, v$

Second-stage decisions depend on  $d$   
i.e.  $y$  and  $z$  are functions of  $d$

# Scenario Tree

Suppose demand  $d$  is a discrete random variable with  $S$  scenarios.



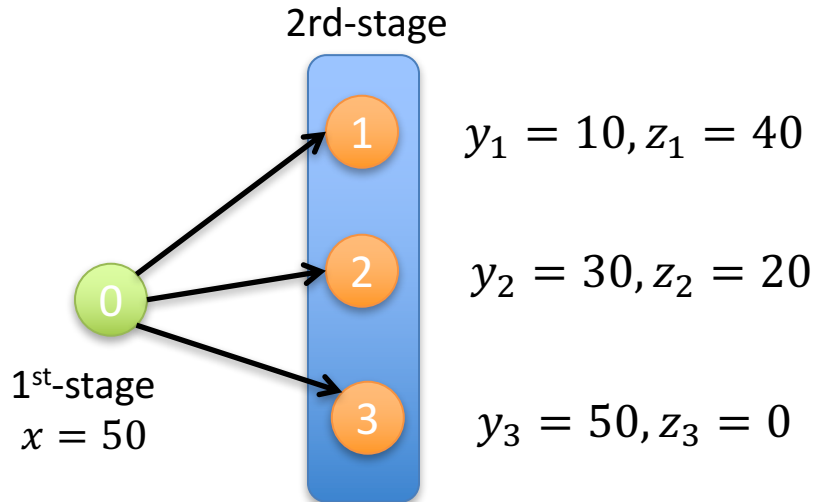
# A Concrete Example

- Suppose the demand  $d$  can be low  $d_1 = 10$ , normal  $d_2 = 30$ , and high  $d_3 = 50$ , with probability  $\frac{1}{4}, \frac{5}{12}, \frac{1}{3}$ .
- Unit cost:  $c = 5$ , Retail price:  $r = 10$ , Discount price:  $s = 3$ .
- Production capacity:  $\bar{x} = 70$ .

$$\begin{aligned} \min_{x, y_i, z_i} \quad & 5x + \frac{1}{4}(-10y_1 - 3z_1) + \frac{5}{12}(-10y_2 - 3z_2) + \frac{1}{3}(-10y_3 - 3z_3) \\ \text{s.t.} \quad & 0 \leq x \leq 70 \\ & y_1 \leq 10, y_2 \leq 30, y_3 \leq 50 \\ & -x + y_i + z_i \leq 0 \quad \forall i = 1, 2, 3 \\ & y_i \geq 0, z_i \geq 0 \quad \forall i = 1, 2, 3 \end{aligned}$$

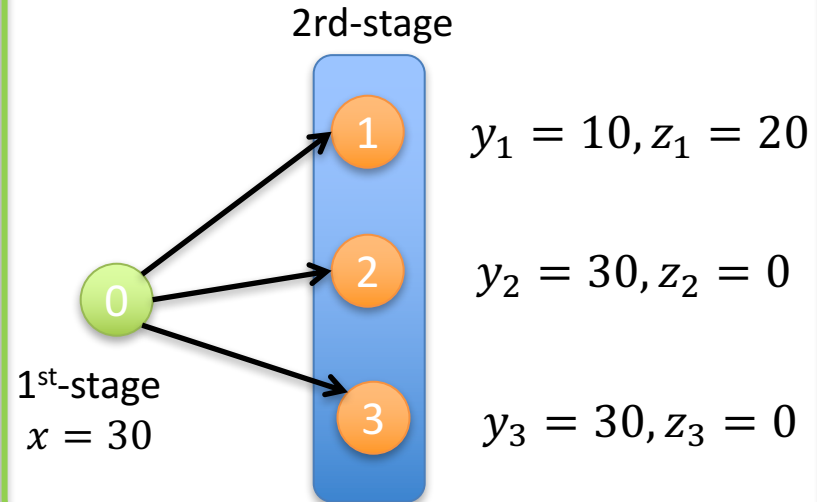
# Result Analysis

- Cost and prices:  $c = 5, r = 10, s = 3$



Solution depends on ratio  $\frac{r-c}{r-s}$

- Cost and prices:  $c = 5, r = 10, s = 2$



# Summary

- We introduced a graphical representation of uncertainty: Scenario tree.
- We solved a concrete example of a two-stage stochastic inventory control problem.
- We analyzed the solution and its dependence on parameters.